Parliament

Project Roadmap & Development Timeline

Version 2.0 | March 2026 | Canada-First Edition

Executive Summary

Parliament is a cross-platform civic technology application designed to inform and politically empower the Canadian voting public. Users enter their postal code and immediately see their elected representatives at every level of government — federal, provincial, and municipal — along with key information about each one.

This version of the roadmap reflects a pivotal strategic decision: build top-down, starting with Canada's 338 federal ridings and all 338 Members of Parliament. This ensures that on launch day, every Canadian — from Vancouver to Halifax — can use the app to find their MP. Ontario provincial coverage (MPPs) and Toronto municipal coverage (City Councillors) follow in subsequent layers, all within Version 1.

The platform is designed for radical scalability from day one. Every data layer — federal, provincial, municipal — feeds into a single unified schema, so adding a new province or city is a matter of connecting a new data source, not rebuilding the application.

Project Overview

Vision

To become the most trusted civic technology platform in the world — giving every citizen a clear, accessible window into their political leadership and a safe, productive space to engage with their fellow citizens across the political divide.

Core Functions

- Civic Information Layer (v1): Representative lookup by postal code at federal, provincial, and municipal levels. Name, party, riding, contact info, date elected, and next election date.
- Civic Discourse Layer (v2+): A structured, polarization-resistant forum where citizens can engage, debate, and build consensus on political issues.

Version 1 Geographic Scope

Version 1 is Canada-focused. Users outside Canada are not addressed in this version. Within Canada, coverage rolls out in three layers:

Layer	Level	Scope	Data Shown
1	**Federal**	All 338 ridings — every Canadian	MP name, party, riding, contact, date elected, next election
2	**Provincial**	Ontario — all 124 ridings (MPPs)	MPP name, party, riding, contact, date elected, next election
3	**Municipal**	Toronto — all 25 wards (Councillors)	Councillor name, ward, party/affiliation, contact, next election

A user in Toronto will see all three layers. A user in Ottawa will see their MP and MPP but no municipal layer (yet). A user in Quebec will see their MP only. Every user gets something useful on day one.

Competitive Differentiation

Existing platforms (X, Facebook, Instagram) serve political information as a byproduct of engagement-maximizing design, which actively harms civic health through polarization and misinformation. Parliament is purpose-built for civic empowerment — prioritizing accuracy, accessibility, and calm over virality and outrage.

Technical Core: The Postal Code District Lookup

The entire application rests on one foundational capability: accept a Canadian postal code as input and return the correct federal riding, provincial riding, and municipal ward for that location. Everything else — displaying representative profiles, voting records, election dates — is a simple database lookup once the districts are known.

The Challenge

Canadian postal codes are assigned by Canada Post for mail delivery, not for electoral purposes. A single postal code can occasionally straddle a riding or ward boundary. The standard solution is to use the geographic centroid (centre point) of the postal code as the lookup coordinate, which is accurate for the vast majority of addresses.

Recommended Approach: Hybrid Lookup

Layer	Method	Data Source	How It Works
FederalRidings	Lookup table	Elections Canada — Postal Code Conversion File (PCCF)	Download Elections Canada's free CSV mapping postal codes to federal electoral districts. Fast, no geometry required.
OntarioRidings	Point-in-polygon	Elections Ontario — Electoral boundary shapefiles	Convert postal code to lat/lng via Nominatim, then check which Ontario riding polygon contains the point.
TorontoWards	Point-in-polygon	City of Toronto Open Data — City Wards dataset	Same lat/lng point checked against Toronto ward boundary polygons from the Open Data portal.

Implementation Stack for the Lookup Service

- Language: Python (beginner-friendly, excellent geospatial libraries)
- Geocoding: Nominatim via OpenStreetMap (free, no API key required, rate-limited)
- Geospatial queries: GeoPandas library (point-in-polygon against shapefiles)
- Federal lookup: Elections Canada PCCF — a flat CSV file, no geometry needed
- This Python service becomes the backend engine called by the app whenever a user enters a postal code

Key Data Sources to Bookmark

Source	What It Provides	URL
Elections Canada PCCF	Postal code → federal riding mapping (CSV)	elections.ca → Electoral Geography
Elections Ontario Boundaries	Ontario riding shapefiles	elections.ontario.ca → Electoral Boundaries
Toronto Open Data — City Wards	Toronto ward boundary shapefiles	open.toronto.ca → search "City Wards"
Nominatim / OpenStreetMap	Free postal code geocoding (lat/lng)	nominatim.openstreetmap.org
Parliament of Canada Open Data	MP profiles, party, riding, contact info	ourcommons.ca → Open Data
Ontario Legislative Assembly	MPP profiles and contact info	ola.org → Members

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Technology Stack

The recommended stack minimizes the number of codebases to maintain while building on skills adjacent to your existing experience (C/C++, HTML/JS, SQL). A single JavaScript codebase covers iOS, Android, and web.

Layer	Technology	Rationale
Frontend (App + Web)	**React Native + React Web (Expo)**	One codebase deploys to iOS, Android, and browser. JavaScript is close to your HTML/JS experience.
Backend / API	**Node.js + Express or Supabase**	JavaScript end-to-end. Supabase provides a free-tier PostgreSQL database with a built-in REST API.
Database	**PostgreSQL (via Supabase)**	Relational DB maps well to your SQL background. Scalable to millions of rows.
District Lookup Service	**Python + GeoPandas + Nominatim**	Handles postal code → riding/ward resolution. Python is beginner-friendly with excellent geospatial tools.
Data Ingestion	**Python scripts (API calls + scraping)**	Fetches and refreshes MP/MPP/councillor data from Elections Canada, Parliament, and city portals.
Hosting	**Vercel (web) + Expo EAS (mobile)**	Both have generous free tiers. Near-zero config deployment.

Version Control	**GitHub**	Essential from day one. Free and industry-standard.

Phased Development Roadmap

Five phases spanning approximately 24–30 months for a solo founder. Timelines compress significantly with collaborators. Each phase has a strict exit milestone — do not advance until it is met.

Phase 1 **Foundation** *Months 1–3*	**Key Tasks** • Learn JavaScript fundamentals and React basics (freeCodeCamp, Scrimba) • Set up GitHub repo, Supabase project, and Expo dev environment • Download Elections Canada PCCF file; build postal code → federal riding lookup in Python • Download Ontario and Toronto boundary shapefiles; build point-in-polygon lookup using GeoPandas • Design scalable DB schema: Country > Province > Riding > Representative • Write Python scripts to fetch all 338 MP profiles from ourcommons.ca Open Data • Verify lookup accuracy: test 20+ postal codes across Canada	**Goals & Milestone** • Postal code → federal riding lookup works for all 338 ridings • Postal code → Ontario riding and Toronto ward lookup works • All 338 MP profiles stored in database **✓ District lookup engine is accurate and fully functional**
Phase 2 **Federal Layer** *Months 4–6*	**Key Tasks** • Build core app UI: postal code input screen, representative profile screen • Wire up postal code input to district lookup → MP profile display • Display for each MP: name, party, riding, contact info, date elected, next election date • Set up automated weekly data refresh (cron job via Supabase or GitHub Actions) • Build responsive web version (React Web via Expo) • Internal testing across diverse Canadian postal codes	**Goals & Milestone** • Any Canadian can enter their postal code and see their MP • App works end-to-end in a web browser • Data refreshes automatically without manual intervention **✓ Federal layer live: every Canadian can find their MP**
Phase 3 **Provincial + Municipal Layers** *Months 7–10*	**Key Tasks** • Fetch all 124 Ontario MPP profiles from the Legislative Assembly of Ontario • Add MPP display layer to app (Ontario users see MP + MPP) • Fetch all 25 Toronto City Councillor profiles from Toronto Open Data • Add councillor display layer (Toronto users see MP + MPP + Councillor) • Build graceful 'layer not yet available' messaging for non-Ontario, non-Toronto users	**Goals & Milestone** • Ontario users see both their MP and MPP • Toronto users see all three levels of representation • Non-Ontario Canadians see their MP with a clear roadmap note for future coverage **✓ All three data layers live — Version 1 feature-complete**

	• UI polish: unified multi-level representative display screen	
Phase 4 **Polish & Launch** *Months 11–14*	**Key Tasks** • User testing with 15–25 real users across Toronto, Ontario, and other provinces • Iterate UI/UX based on feedback • Build native mobile app (iOS + Android via Expo/React Native) • Submit to Apple App Store and Google Play • Set up privacy-respecting analytics (PostHog or Plausible) • Launch landing page; grassroots outreach to civic orgs, Reddit, local media • Begin grant applications (Mozilla, Knight Foundation, civic tech funds)	**Goals & Milestone** • App live on iOS, Android, and web • Positive testimonials from real users across multiple provinces • Media or civic organization coverage secured • 100+ active users in first month **✓ Version 1.0 publicly launched Canada-wide**
Phase 5 **Scale + Discourse** *Months 15–30+*	**Key Tasks** • Add additional provinces (BC, Alberta, Quebec) — provincial layer expansion • Add additional municipalities beyond Toronto • Design and prototype the structured civic discussion feature (v2) • Research and implement polarization-reduction mechanisms • Build moderation tools (AI-assisted + human moderator support) • Explore sustainable funding: grants, philanthropy, B2G partnerships • Evaluate international expansion (UK, USA, Australia)	**Goals & Milestone** • Provincial coverage expanded beyond Ontario • Discourse feature in public beta • Sustainable funding model identified **✓ Version 2.0 with civic discourse feature launched**

Timeline at a Glance

Phase	Timeline	Key Deliverable	Exit Milestone
1 — Foundation	Months 1–3	District lookup engine	Postal code correctly resolves all three layers
2 — Federal Layer	Months 4–6	Federal web prototype	Any Canadian finds their MP
3 — Prov. + Municipal	Months 7–10	All three layers in-app	Toronto users see MP + MPP + Councillor
4 — Launch	Months 11–14	iOS + Android + Web live	V1.0 public launch Canada-wide
5 — Scale + Discourse	Months 15–30+	More provinces + V2 discourse	V2.0 with civic discussion in beta

Risk Register

Risk	Likelihood	Impact	Mitigation
Postal code boundary ambiguity — a code straddles two ridings	**Low**	**Low**	Use centroid-based lookup. For genuine boundary cases, show both ridings and let the user confirm their address.
Elections Canada PCCF file is outdated after a redistribution	**Medium**	**High**	Monitor Elections Canada announcements. Subscribe to their data update feeds. Re-download and re-validate after any redistribution.
ourcommons.ca Open Data structure changes, breaking the MP data pipeline	**Medium**	**Medium**	Build data ingestion defensively with error alerting. Check pipeline output weekly.
Scope creep — adding provinces before v1 is solid	**High**	**Medium**	Strict phase gates. Do not begin Phase 5 provincial expansion until v1.0 is publicly live.
Solo developer burnout slows progress significantly	**High**	**High**	Set sustainable weekly hour commitments. Actively recruit a technical co-founder or volunteer contributor by Month 6.
Political bias perception damages trust	**Medium**	**High**	Data-only, never editorial. Transparent sourcing. Advisory board with diverse political representation from day one.
App Store rejection (politics-adjacent content)	**Low**	**Medium**	Frame as civic information tool. Review Apple and Google civic app guidelines before submission.

Recommended Learning Path

Given your background (C/C++ via Arduino, basic HTML/JS, SQL), this sequence builds the skills needed for Phase 1 and 2 without overwhelming you. Your existing knowledge transfers more than you may think.

Months 1–2: Core Web Skills

- **Track:** JavaScript fundamentals — freeCodeCamp.org, JavaScript Algorithms and Data Structures (free). Complete the first 100 lessons.
- **Track:** React basics — Scrimba React course (free tier) or The Odin Project.
- **Track:** Git & GitHub — GitHub's "Git & GitHub for Beginners" on YouTube (1 hour). Set up your repo on Day 1.

Month 2–3: Python & Data

- **Track:** Python for data work — Automate the Boring Stuff with Python (free at automatetheboringstuff.com). Focus on file handling and web requests.
- **Track:** GeoPandas for geospatial lookups — GeoPandas official docs + YouTube tutorials. Your C++ logic skills make this approachable.
- **Track:** SQL refresher + Supabase — Supabase official docs. Your existing SQL knowledge makes this fast.

Ongoing: Build in Public

The fastest way to learn is to build. Commit code to GitHub daily, even small changes. Consider documenting your journey publicly — this builds an early audience, attracts collaborators, and strengthens future grant applications.

Funding & Sustainability Strategy

Bootstrap Phase (v1 — Months 1–14)

Estimated monthly infrastructure cost: under $50/month. Supabase, Vercel, and Expo all have generous free tiers that comfortably cover v1 traffic. A domain name costs approximately $15/year. Personal funding is entirely feasible for this phase.

Grant Opportunities (Phase 4 onward)

- Mozilla Foundation — Civic Tech and Internet Health grants
- Knight Foundation — journalism and civic information funding (US-based but open to Canadian projects)
- Innovation, Science and Economic Development Canada (ISED) — digital infrastructure programs
- Journalism school partnerships — Toronto Metropolitan University, University of Toronto
- Toronto Community Foundation — civic innovation grants

Long-Term Sustainability

As a registered not-for-profit, Parliament can pursue a mixed model: institutional grants, municipal B2G partnerships, and optional premium tools for civic organizations and journalism partners. Advertising is explicitly not part of the model.

Immediate Next Steps (This Week)

#	Action	Resource
1	Create a GitHub account and a repo named ‘Parliament’	github.com
2	Download the Elections Canada Postal Code Conversion File (PCCF)	elections.ca → Electoral Geography
3	Download Toronto ward boundary shapefiles from Toronto Open Data	open.toronto.ca → “City Wards”
4	Download Ontario riding boundary shapefiles from Elections Ontario	elections.ontario.ca → Electoral Boundaries
5	Install Python, VS Code, and the GeoPandas library on your computer	python.org / code.visualstudio.com
6	Create a free Supabase account and initialize a project	supabase.com
7	Begin freeCodeCamp JavaScript course (first 5 lessons)	freecodecamp.org

“Every Canadian deserves to know who represents them. You’re building that.”

This roadmap will evolve. What matters is starting.